

# Story Animator

Project Team

|                    |          |
|--------------------|----------|
| Noor Ul Huda       | 21I-1357 |
| Farhan Javaid      | 21I-1671 |
| Mohammad Shaheryar | 21I-0776 |

Session 2021-2025

Supervised by

Ms. Ayesha Kamran Ul Haq



Department of Computer Science

National University of Computer and Emerging Sciences  
Islamabad, Pakistan

October, 2024

# Contents

|                                                      |    |
|------------------------------------------------------|----|
| 1 Introduction .....                                 | 1  |
| 1.1 Problem Statement.....                           | 1  |
| 1.2 Scope .....                                      | 1  |
| 1.2.1 Core Functionality.....                        | 1  |
| 1.3 Modules.....                                     | 2  |
| 1.3.1 Image Generation and Modification .....        | 2  |
| 1.3.2 AI Character Generator .....                   | 2  |
| 1.3.3 Easy GIF Creation .....                        | 2  |
| 1.4 User Classes and Characteristics.....            | 3  |
| 2 Project Requirements .....                         | 5  |
| 2.1 Use-case/Event Response Table/Storyboarding..... | 5  |
| 2.2 Functional Requirements.....                     | 8  |
| 2.2.1 User Authentication .....                      | 9  |
| 2.2.2 Image Generation .....                         | 9  |
| 2.2.3 Convert Image to SVG .....                     | 9  |
| 2.2.4 Background Removal/Noise Reduction .....       | 10 |
| 2.2.5 GIF Generation .....                           | 10 |
| 2.2.6 Manage Chatbot History.....                    | 10 |
| 2.3 Non-Functional Requirements .....                | 11 |
| 2.3.1 User Experience .....                          | 11 |
| 2.3.2 Industry Impact.....                           | 11 |
| 2.3.3 Reliability .....                              | 11 |
| 2.3.4 Security .....                                 | 11 |
| 2.3.5 Scalability .....                              | 11 |
| 3 System Overview .....                              | 12 |
| 3.1 Architectural Design .....                       | 12 |
| 3.2 Design Models.....                               | 13 |

|                                                |    |
|------------------------------------------------|----|
| 3.2.1 Activity Diagram.....                    | 13 |
| 3.2.2 Class Diagram .....                      | 15 |
| 3.2.3 Class-level Sequence Diagrams.....       | 16 |
| 3.2.3.1 User Sign-up Process.....              | 16 |
| 3.2.3.2 User Login Process .....               | 19 |
| 3.2.3.3 Image Upload Process.....              | 19 |
| 3.2.3.4 GIF Generation Process.....            | 20 |
| 3.2.3.5 Image Download Process .....           | 20 |
| 3.2.3.6 Image to SVG Conversion Process.....   | 21 |
| 3.2.3.7 Background/Noise Removal Process.....  | 21 |
| 3.2.3.8 Conversation History Management.....   | 22 |
| 3.2.4 System-level Sequence Diagrams .....     | 23 |
| 3.2.4.1 User Sign-up Process .....             | 23 |
| 3.2.4.2 User Login Process .....               | 24 |
| 3.2.4.3 AI Image Generation Process .....      | 25 |
| 3.2.4.4 Image Upload Process .....             | 26 |
| 3.2.4.5 Background/Noise Removal Process ..... | 27 |
| 3.2.4.6 Image to SVG Conversion Process .....  | 28 |
| 3.2.4.7 Image Download Process .....           | 29 |
| 3.2.4.8 Chat History Management Process .....  | 30 |
| 3.3 Data Design .....                          | 31 |
| 3.3.1 Data Flow Diagram .....                  | 31 |
| References .....                               | 33 |

# List of Figures

|      |                                                 |    |
|------|-------------------------------------------------|----|
| 2.1  | Usecase Diagram .....                           | 6  |
| 2.2  | Tracing Issues .....                            | 11 |
| 3.1  | Architectural Diagram .....                     | 13 |
| 3.2  | Activity diagram .....                          | 15 |
| 3.3  | Class Diagram .....                             | 17 |
| 3.4  | User Sign-up Process Sequence .....             | 18 |
| 3.5  | User Login Process Sequence .....               | 19 |
| 3.6  | Image Upload Process Sequence .....             | 19 |
| 3.7  | GIF Generation Process Sequence .....           | 20 |
| 3.8  | Image Download Process Sequence .....           | 20 |
| 3.9  | Image to SVG Conversion Sequence .....          | 21 |
| 3.10 | Background/Noise Removal Sequence .....         | 21 |
| 3.11 | chat history management .....                   | 22 |
| 3.12 | User Sign-up Process Sequence .....             | 23 |
| 3.13 | User Login Process Sequence .....               | 24 |
| 3.14 | AI Image Generation Process Sequence .....      | 25 |
| 3.15 | Image Upload Process Sequence .....             | 26 |
| 3.16 | Background/Noise Removal Process Sequence ..... | 27 |
| 3.17 | Image to SVG Conversion Process Sequence .....  | 28 |
| 3.18 | Image Download Process Sequence .....           | 29 |
| 3.19 | Chat History Management Process Sequence .....  | 30 |
| 3.20 | Data Flow Diagram .....                         | 31 |



# List of Tables

|                                                      |   |
|------------------------------------------------------|---|
| 1.1User Class .....                                  | 4 |
| 2.1Use Case UC1 - Login .....                        | 7 |
| 2.2Use Case UC2 - Signup .....                       | 7 |
| 2.3Use Case UC3 - Generate Image from Text .....     | 7 |
| 2.4Use Case UC4 - Import Own Image .....             | 8 |
| 2.5Use Case UC5 - Convert Image to SVG .....         | 8 |
| 2.6Use Case UC6 - Download Image .....               | 8 |
| 2.7Use Case UC7 - Remove Background/Noise .....      | 9 |
| 2.8Use Case UC8 - Generate GIF from Text .....       | 9 |
| 2.9 Use Case UC9 - Manage Chatbot History (DB) ..... | 9 |

# Chapter 1

## Introduction

Available animated characters are fairly common in present day's digital setting. It is possible to meet them in TV series, motion pictures and video games and even in social media ads. But to create these characters is rather impossible. It involves a lot of money, time and even skills for its success to be achieved. Here come the story animator, the project that we have developed to address this issue. That is why we are creating a new tool which will significantly ease the process of character creation with the help of artificial intelligence. To both those who need an animator to draw with for them to earn a living and to those who just draw because they love it. It perform many of the timeconsuming operations on its own

### 1.1 Problem Statement

Creating characters and modifying them pose great challenges to the animation industry. Creativity is slightly restricted, prices are high, and the working process takes a lot of time for designers and animation businesses. Working longer hours working from the scratch and employing more designers translate to high costs to the businesses. This can make it more difficult to explore new ideas and decrease the time needed to develop a character concept and design. As a result, it is challenging for the design and animation sectors to promote innovation and operate effectively.

### 1.2 Scope

The current project involves the development of a chatbot web application with a vision to revolutionize animation and character design. Through using ai image and GIF generation in conjunction with creative tools, we have a platform suitable for the amateur artist as

1. Introduction

well as the professional one.

#### 1.2.1 Core Functionality

key features include:

- 
- • Automatic object tracking and separation
  - • Image to SVG conversion
  - • AI-powered background removal
  - • AI character generation
  - • Story-based animation prompts
  - • Gif motion generation

## 1.3 Modules

Story Animator functionality is organized into the following core modules:

### 1.3.1 Image Generation and Modification

This module forms the backbone of our character design toolset. In this module will generate AI character image and then we will use object tracing and separation techniques for easy manipulation of character parts. We will offer background removal, giving users more control over final appearance.

1. Object Tracing and Separation
2. SVG Conversion
3. Background Removal

### 1.3.2 AI Character Generator

Taking an advantage of AI's creative prowess in conception of characters and its time saving aspect in characterization.

1. Text-to-Image Feature
2. Style Options

### 1.4 User Classes and Characteristics

### 1.3.3 Easy GIF Creation

Simplify the process of bringing static designs to life. Convert static character designs into simple animated GIFs with minimal user effort.



- GIF Generation

## 1.4 User Classes and Characteristics

Our product can be used by multiple people:

1. Animation studios and production companies
2. Independent animators and character designers
3. Graphic designers and illustrators
4. Content creators and social media influencers
5. Advertising and marketing agencies
6. Game development studios

| User class                        | Description                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Animation studios                 | Using our chatbot animation studio or product companies don't have to generate there characters or images from scratch they can use our image generation feature and generate the image. Modify the image according to their own use.                                                                                      |
| Animators and Character designers | Animators can generate characters. Then download that image in a svg format so they can easily redesign that image according to their own need. They can use background removal feature so they have more control over final appearance. GIF generation can be used by animators to bring life to their static characters. |
| Content creators                  | Content Creators can make small videos from our chatbot. They can make multiple small videos of 2-3 secs and then merge these videos to make their reels or story.                                                                                                                                                         |
| Game development studios          | Game developers can generate Sprite characters for their games and use GIF generation for the movement of sprites                                                                                                                                                                                                          |

Table 1.1: User Class

## Chapter 2

# Project Requirements

## 2.1 Use-case/Event Response Table/Storyboarding

In this section we will discuss use case Diagram and show use case table. In our project firstly user will login or sign up or web based chat bot. Then user can generate an AI image. User can use that image further and convert it into editable format or remove background/noise. User can import his/her own image and manipulate its image. User can download the image and further change it to his needs. One of the main features of our chat bot is GIF generation. User can generate GIF using image and textual prompt. Following is the use case Diagram and table:

## 2. Project Requirements

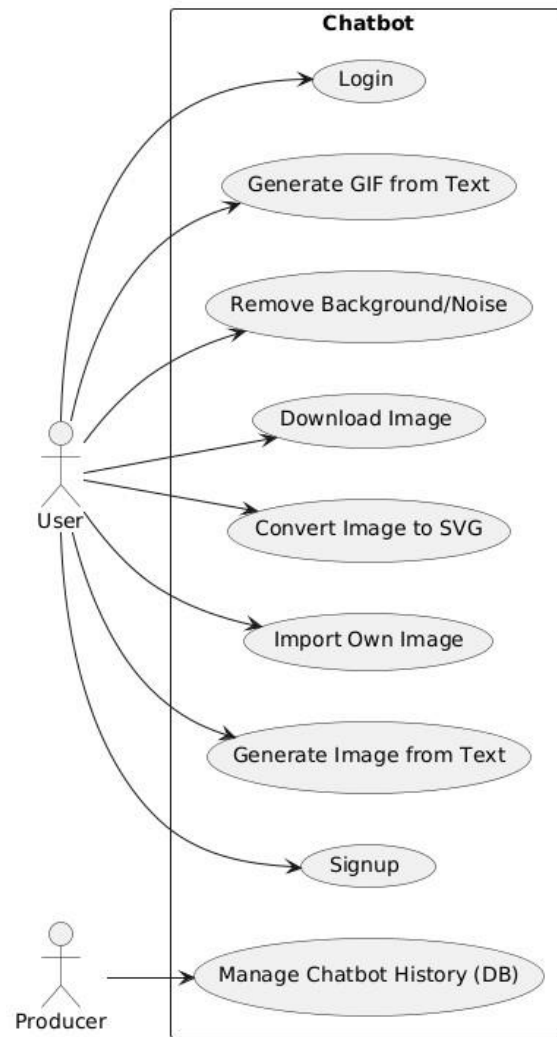


Figure 2.1: Usecase Diagram

### 2.1 Use-case/Event Response Table/Storyboarding

|             |                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID | UC1                                                                                                                                                                                                                                                                                                                                                                                                       |
| Name        | Login                                                                                                                                                                                                                                                                                                                                                                                                     |
| Description | This use case allows registered users to securely log into the chatbot platform. The user will be required to enter valid credentials (such as username and password). If the credentials match the stored records, the system grants access. Otherwise, the user will be prompted with an error message and asked to retry. Multi-factor authentication (MFA) may also be integrated for added security. |
| Actors      | User                                                                                                                                                                                                                                                                                                                                                                                                      |

Table 2.1: Use Case UC1 - Login

|             |                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID | UC2                                                                                                                                                                                                                                                                                                                                                                          |
| Name        | Signup                                                                                                                                                                                                                                                                                                                                                                       |
| Description | Through this use case, new users can register for a platform account. Users will be prompted to enter any relevant information, including their password, email address, and phone number, etc. After creating an account successfully, an email verification procedure might be used to confirm the account. By doing this, the platform may track users for future logins. |
| Actors      | User                                                                                                                                                                                                                                                                                                                                                                         |

Table 2.2: Use Case UC2 - Signup

|             |                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID | UC3                                                                                                                                                                                                                                                                                                                                                                                                        |
| Name        | Generate Image from Text                                                                                                                                                                                                                                                                                                                                                                                   |
| Description | In this use case, the user inputs a textual prompt or description, and the system uses the AI model's interpretation of the prompt to create an image. The meaning, tone, or keywords in the supplied text will be reflected in the created image. Users can refine their inputs to provide more precise or targeted outcomes. This procedure turns text into visual art by using machine learning models. |
| Actors      | User                                                                                                                                                                                                                                                                                                                                                                                                       |

Table 2.3: Use Case UC3 - Generate Image from Text

|             |                                                                                                                                                                                                                                                                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID | UC4                                                                                                                                                                                                                                                                                                                                                         |
| Name        | Import Own Image                                                                                                                                                                                                                                                                                                                                            |
| Description | Using this case, users can contribute with their own pictures to the platform for further enhancements or editing. The proposed system should, however, support more file formats other than the aforementioned namely; PNG, JPEG, and BMP. Background removal and image format converter can be used by the user after the upload of the particular image. |
| Actors      | User                                                                                                                                                                                                                                                                                                                                                        |

Table 2.4: Use Case UC4 - Import Own Image

|             |                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID | UC5                                                                                                                                                                                                                                                                                                                                                                           |
| Name        | Convert Image to SVG                                                                                                                                                                                                                                                                                                                                                          |
| Description | In this particular use case, the user is able to transform a picture that has been created or copied to the SVG (Scalable Vector Graphic) format within use of the system. This conversion is ideal for logos, icons, as well as for illustrations. As a result, such users are free to download the SVG file or use more features of the platform for further customization. |
| Actors      | User                                                                                                                                                                                                                                                                                                                                                                          |

Table 2.5: Use Case UC5 - Convert Image to SVG

|             |                                                                                                            |
|-------------|------------------------------------------------------------------------------------------------------------|
| Use Case ID | UC6                                                                                                        |
| Name        | Download Image                                                                                             |
| Description | With this use case, users can download a picture that they have modified after generating or importing it. |
| Actors      | User                                                                                                       |

Table 2.6: Use Case UC6 - Download Image

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID | UC7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Name        | Remove Background/Noise                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Description | This use case enables users to remove off the noise in an image, meaning that only the necessary portions of the image will be left. It gives users a choice to use an AI-created image or upload their own of choice to the app. It has features that allow it to eliminate background as a possibility and keep subject of interest as priority. This function is useful to pick up necessary items, to increase the transparency of the background, and to increase overall sharpness of the picture. |
| Actors      | User                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Table 2.7: Use Case UC7 - Remove Background/Noise

|             |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID | UC8                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Name        | Generate GIF from Text                                                                                                                                                                                                                                                                                                                                                                                                       |
| Description | In this use case, the system creates a GIF in response to the prompt after the user enters a text description that includes dynamic elements (like verbs). The AI interprets the movements and actions mentioned in the text and converts them into animated graphics, resulting in a brief image loop that is then converted into a GIF. The creation of original and compelling material is made possible by this feature. |
| Actors      | User                                                                                                                                                                                                                                                                                                                                                                                                                         |

Table 2.8: Use Case UC8 - Generate GIF from Text

|             |                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID | UC9                                                                                                                                                                                                                                                                                                                                                                                                   |
| Name        | Manage Chatbot History (DB)                                                                                                                                                                                                                                                                                                                                                                           |
| Description | Database administration is the main use case, which handles chatbot conversation history. The producer, also known as the backend system, keeps a database copy of the user's conversation history so that it may be retrieved, viewed, and edited as needed. Because past exchanges can be consulted and utilized to improve current discussions, this enables a more customized chatbot experience. |
| Actors      | Producer                                                                                                                                                                                                                                                                                                                                                                                              |

Table 2.9: Use Case UC9 - Manage Chatbot History (DB)

## 2.2 Functional Requirements

The Functional requirement of our project are as follows:

- User Authentication
- Image Generation
- Convert Image to SVG
- Background Removal/Noise Reduction
- GIF Generation
- Manage Chatbot History

---

### 2.2.1 User Authentication

Following are the requirements for module 1:

1. A login page will be available where existing users will login
2. A signup page will be available where new users will make an account

### 2.2.2 Image Generation

To solve our image generation task, we test multiple models to get cartoon characters according to our use case. Some models miserably failed to generate image with a high quality(greater than 1080x720p), while some models are giving noise in output. Here are some models we test:

1. Runwayml/stable-diffusion-v1-5: Output of v1-5 optimised around 512x512 pixels, that's the major issue we're having with it.
2. Stabilityai/stable-diffusion-2-1: Results become so messed up and high contrasted, mostly cartoonized in a bad way
3. Playgroundai/playground-v2.5

Finally we've decide playgroundai version 2.5 ust because of it's particular focus on enhanced color and contrast, improved generation for multi-aspect ratios, and improved human-centric fine details.

### 2.2.3 Convert Image to SVG

To make image in an editable form we will convert our image into svg format using using tracing and contouring techniques. Here are some of the techniques we tried in this module and some issues faced:

- Loss of color information: Tracing typically converts the image to grayscale or binary, losing all color information from the original RGB image. This results in a monochrome output that doesn't capture the richness of the original image.
- Edge detection limitation: Tracing relies heavily on edge detection, by differentiating between colors inside images. But when similar contrast were found tracing could not make a boundary

2.2 Functional Requirements

---

## 2. Project Requirements



Figure 2.2: Tracing Issues

So we decided to search other techniques that would help us to resolve the tracing problem. Then after understanding image processing in depth, we got introduced with a Contours.

What actually Contour is? A curve joining all the continuous points (along the boundary), having same color or intensity. The contours are useful for shape analysis and object detection and recognition.

So we got a perfect boundary detection in contouring along with tracing.

### 2.2.4 Background Removal/Noise Reduction

For removing noise from background and uniquely identify a character from an image, we look forward to Python image processing. But the implementation of kernel results in loss of image quality. To address the issue, we move towards BRIA Background Removal which suddenly help us to effectively separate foreground from background in a range of categories and image types.

### 2.2.5 GIF Generation

With this module, users can combine a text prompt with an image that has been uploaded or created to make animated GIFs. The system's primary task will be to recognize verbs or actions in the text and convert them into lively, dynamic sequences. With the help of this tool, users may create dynamic and captivating content from static photos that can be shared on several platforms. After analyzing the textual input, the AI will create animations using user-defined elements and provide GIF customizing choices.

### 2.2.6 Manage Chatbot History

The management and archiving of users' previous chatbot interactions falls under the purview of this module. The platform can effectively retain discussion logs and guarantee their easy retrieval for subsequent sessions by utilizing MongoDB, a NoSQL database that is both versatile and scalable. Because of its ability to remember past



---

exchanges, this enables a more customized chatbot experience, allowing users to continue where they left off and improving the chatbot's overall functionality.

## 2.3 Non-Functional Requirements

### 2.3.1 User Experience

Through the development of our platform, it becomes easy for users to be able to create animated GIFs which in turn enhances their ability to transform their designs or even subjective stories. Our chatbot is capable of opening a normal bitmap image file; in addition, it can be imported to be used with Adobe Illustrator so it could be used by both beginners and advanced users.

### 2.3.2 Industry Impact

Thus, our project aims to significantly reduce the character creation's time and resources consumed while focusing on these characteristics and enhancing user-friendliness. By story animator the whole animation industry could actually have its ways of design and production altered for the better, enabling people to get the character designs of high quality at reasonable prices.

### 2.3.3 Reliability

- Our chatbot would guarantee availability for consumers by maintaining an uptime of at least 99
- Unexpected mistakes would be graciously handled by the system.

### 2.3.4 Security

- The technology would protect users' personal information, such as usernames and passwords.

### 2.3.5 Scalability

- When additional people or more complicated duties (like picture modification) are added, the system will be able to handle the added demand.

# Chapter 3

## System Overview

Give a general description of the functionality, context, and design of your project. Provide any background information if necessary.

### 3.1 Architectural Design

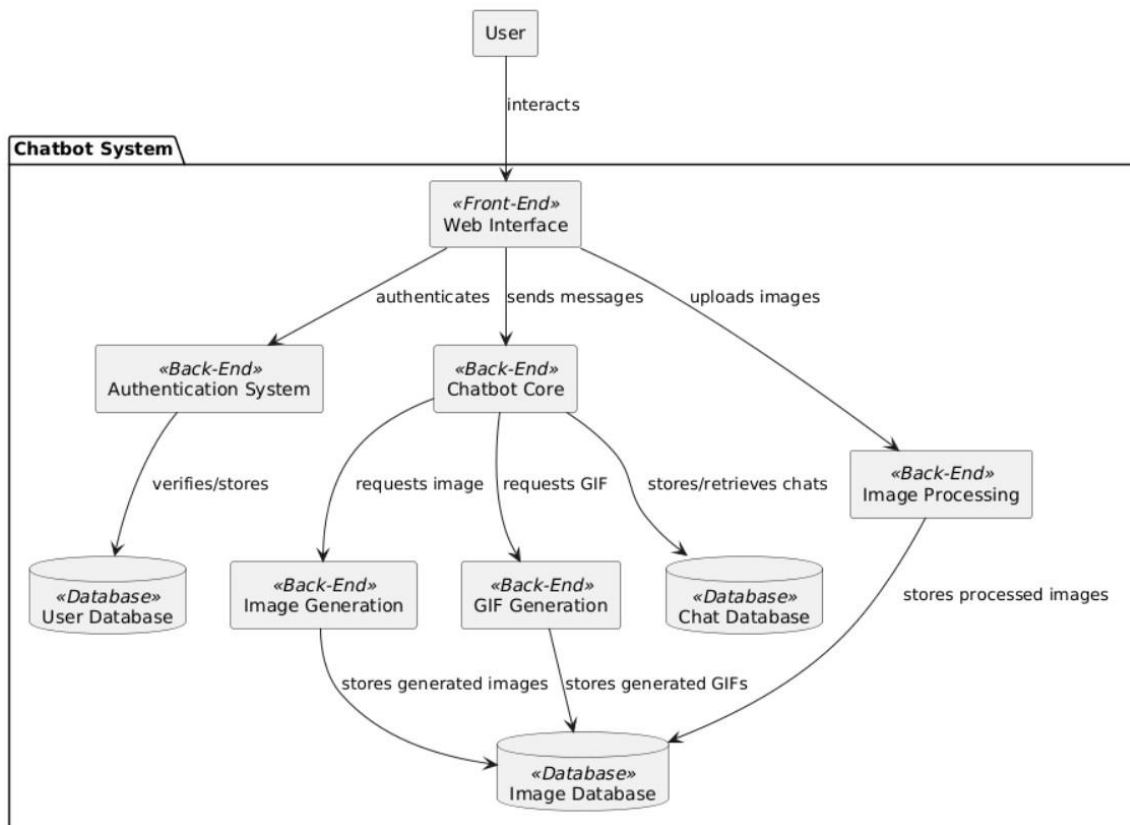


Figure 3.1: Architectural Diagram

The architectural diagram (Figure 3.1) illustrates the structure of our chatbot system. It consists of the following main components:

- User Interface: A web-based front-end that users interact with directly.
- Authentication System: Manages user login and registration, interfacing with the User Database.
- Chatbot Core: Maintaining the context of image for new image generation
- Image Generation: text to image generation along with background removal.
- Image Processing: Conversion of raster images into svg format.

- GIF Generation: On user prompt he/she can easily generate gif videos.
- Databases: To store user chat history along with images, we'll use the database.

Our architecture allows scalability along with the easy separation of concerns between system components.

## 3.2 Design Models

Basically we are using Object-Oriented Development approach which involves:

### 3.2.1 Activity Diagram

This is how the activity flow of the user is looked like in our project. First, if the user of the web site is already registered, log into personal account is to be made. They will make one if not. Following that, customers can upload one image from the computer or create it. A few options will be available to the user, such as a download option, the ability to break up an object (SVG transform option), or making a GIF..

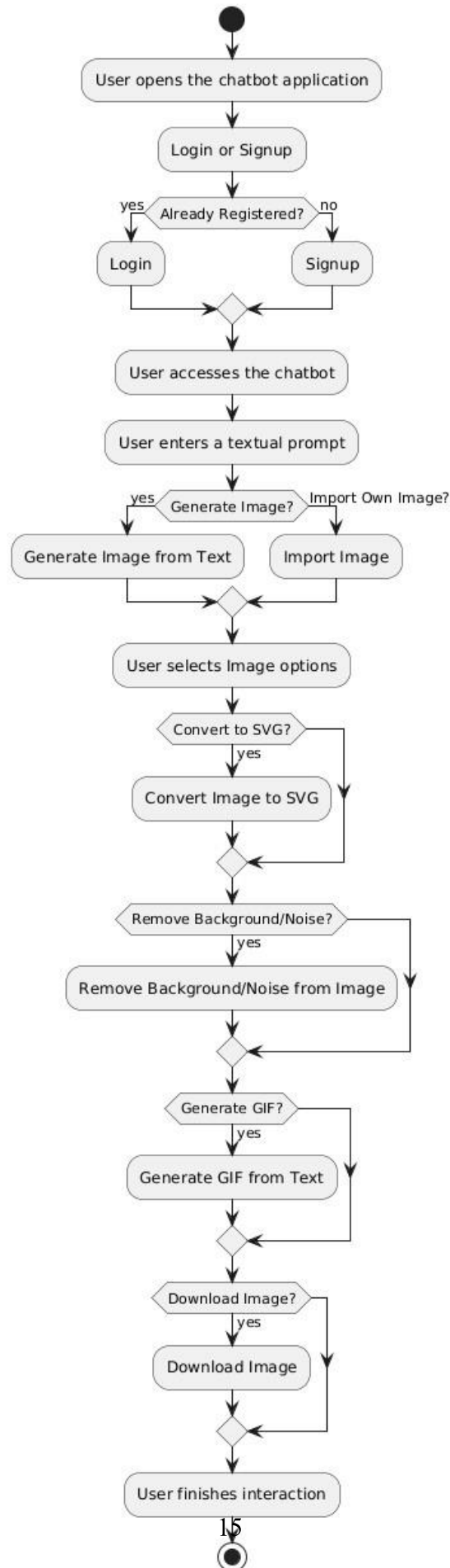


Figure 3.2: Activity diagram

### 3.2.2 Class Diagram

Image representing class diagram, which is showing the relationships among components in our chatbot system.

- **User:** The class having attributes i.e. username, email, and password, and methods like login(), signup(), and useChatbot(), allows user to interact with the platform.
- **Auth:** This class is specialized for handling authentication with methods like LoginPage(), signupPage(), and validateCredentials().
- **Chatbot:** Contains methods for generating images, SVG conversion, removal of noise from backgrounds, generating GIFs, and assisting various user categories (e.g., animators, designers). The Chatbot interacts with both the User and Auth classes.
- **Image:** Stores image data like format, size, and content, with methods to convert, remove background, denoise, and download the image.
- **GIF:** Manages GIF-specific functionality with attributes like frameRate and methods to generate GIFs from text and download them.

### 3. System Overview

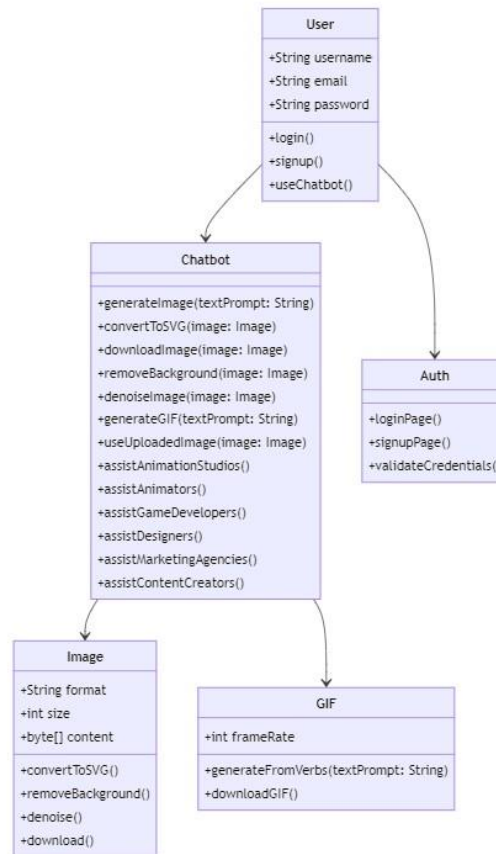


Figure 3.3: Class Diagram

#### 3.2.3 Class-level Sequence Diagrams

##### 3.2.3.1 User Sign-up Process

This diagram is showcasing the process of user registration, including username availability check and account creation.

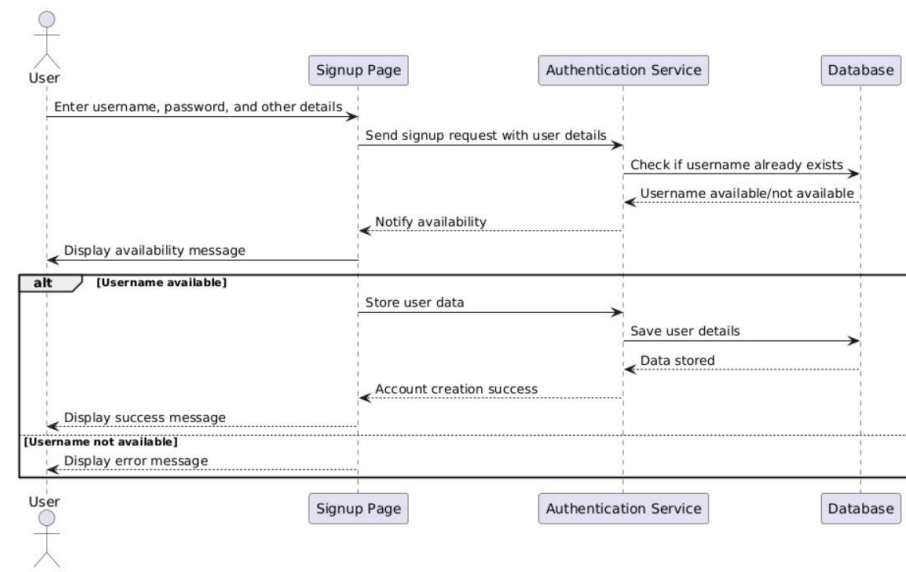


Figure 3.4: User Sign-up Process Sequence

### 3. System Overview

#### 3.2.3.2 User Login Process

Following sequence diagram is showing the login process, including credential validation and JWT token generation.

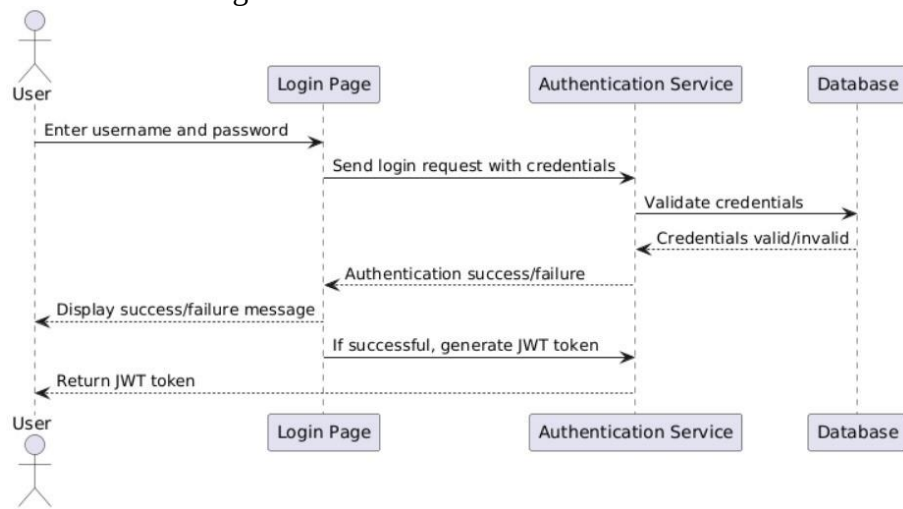


Figure 3.5: User Login Process Sequence

#### 3.2.3.3 Image Upload Process

This diagram shows the basic flow of uploading an image through the user interface and storing them into a database.

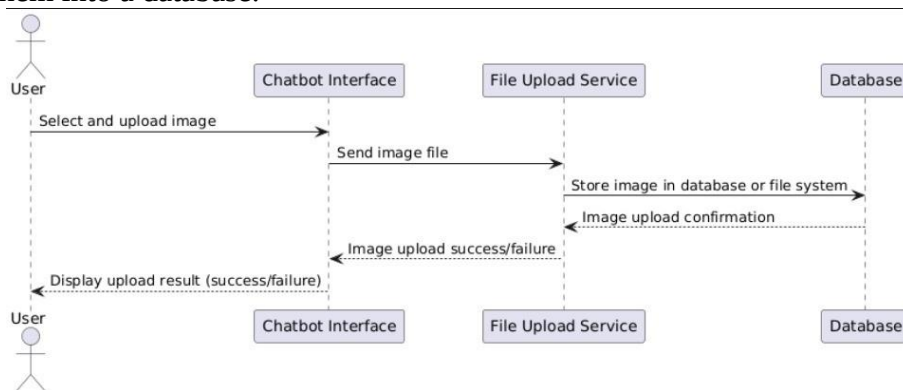


Figure 3.6: Image Upload Process Sequence



### 3.2.3.4 GIF Generation Process

Basically user have to option either to enter a prompt for gif generation, either he/she want's our chatbot to generate gif than chatbot can create a storyline according to the image by using detection models and from them our large language model extract verbs and on basis of that GIF is created.

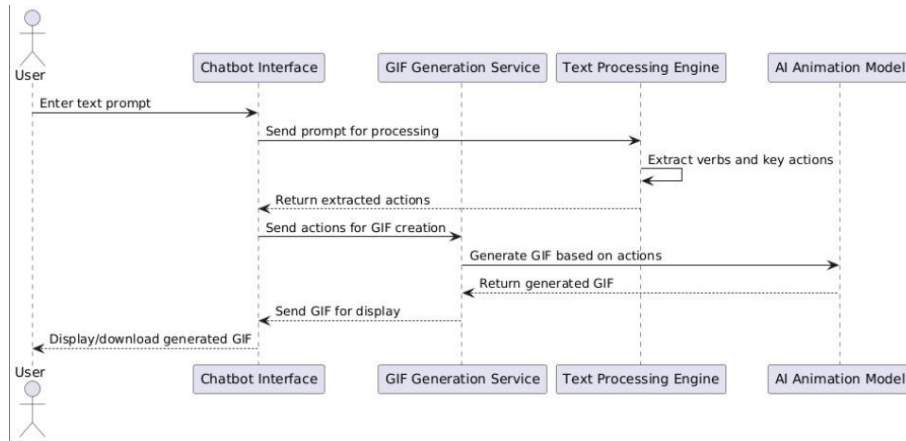


Figure 3.7: GIF Generation Process Sequence

### 3.2.3.5 Image Download Process

User can hover to the image and option of download is shown, than user click on download to get it on local machine.

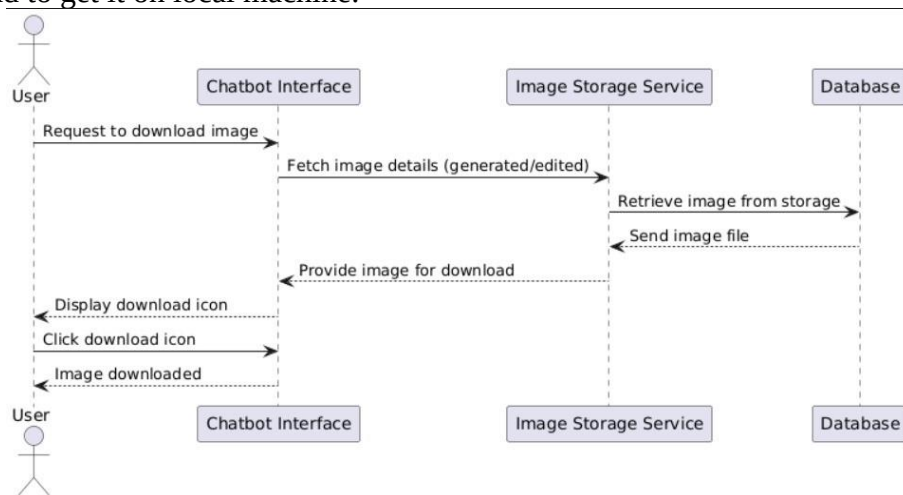


Figure 3.8: Image Download Process Sequence

#### 3.2.3.6 Image to SVG Conversion Process

Upon hovering on an image, user may convert it into SVG format by clicking on the option.

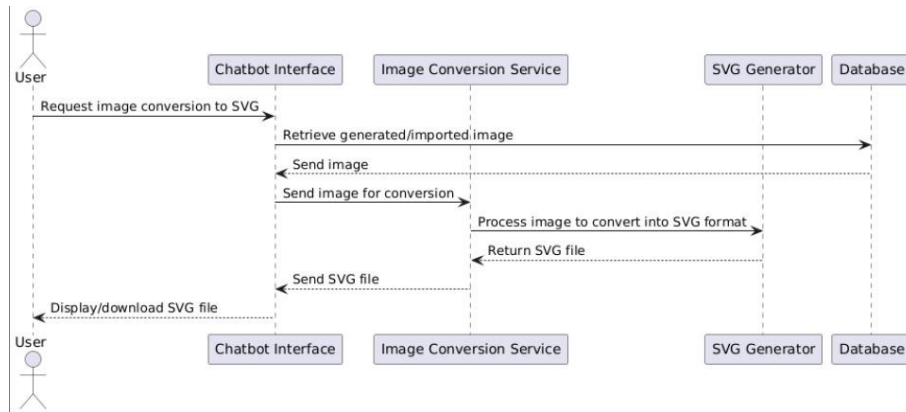


Figure 3.9: Image to SVG Conversion Sequence

#### 3.2.3.7 Background/Noise Removal Process

AI generated images are not perfect ever, they may have some noise in the background. To detect smartly and remove them, we provide a option to user so he/she can easily remove the noise.

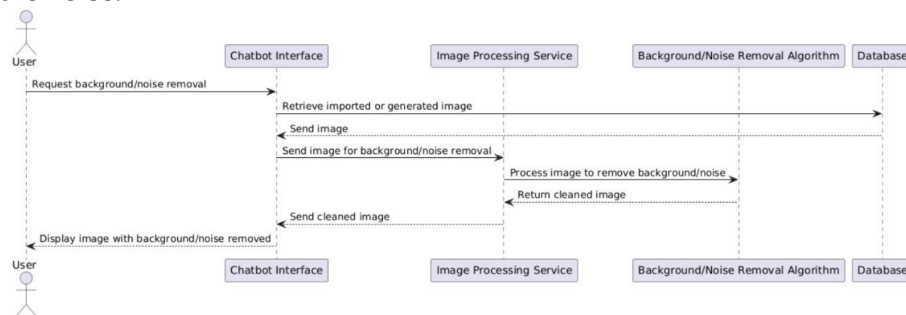


Figure 3.10: Background/Noise Removal Sequence

### 3.2.3.8 Conversation History Management

While user is having conversation with chatbot, we smartly storing all the data to database so if in future user want to continue the chat, he/she have no worries regarding to that and all the chat along with images will be available to them.

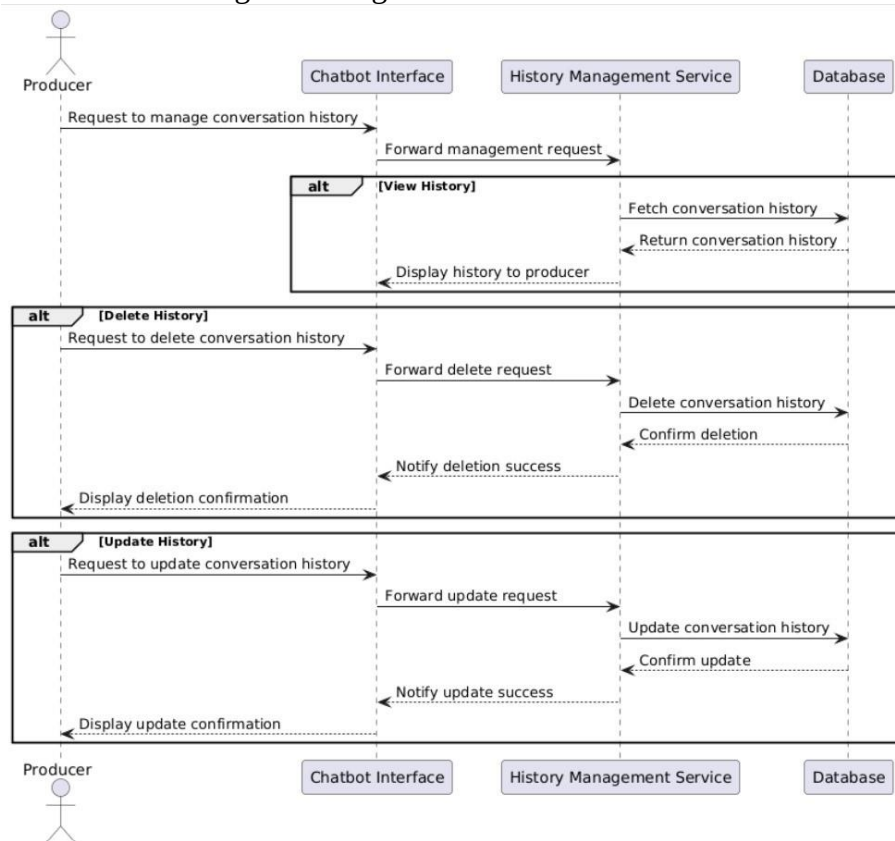


Figure 3.11: chat history management

### 3.2.4 System-level Sequence Diagrams

Following layout provide a brief view of how user interacts with various system components like flow of the operations for key functions.

#### 3.2.4.1 User Sign-up Process

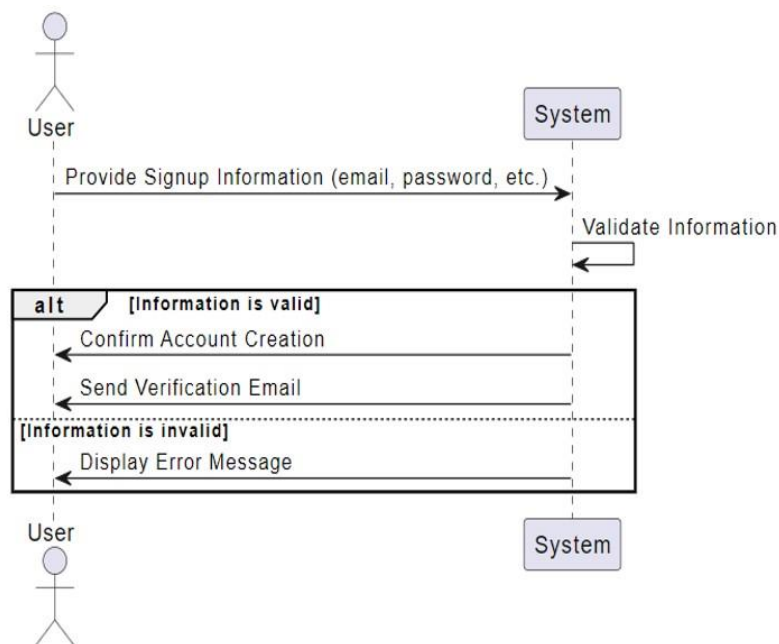


Figure 3.12: User Sign-up Process Sequence

The sign-up process involves:

- User have to enter signup credentials
- System validates input across database if he/she exists or not
- If the user doesn't exist, than system hashes the password so in case network breaches information still be encrypted
- Successfully account is created and user receives a success message
- If not the case, error is thrown

## 3.2.4.2 User Login Process

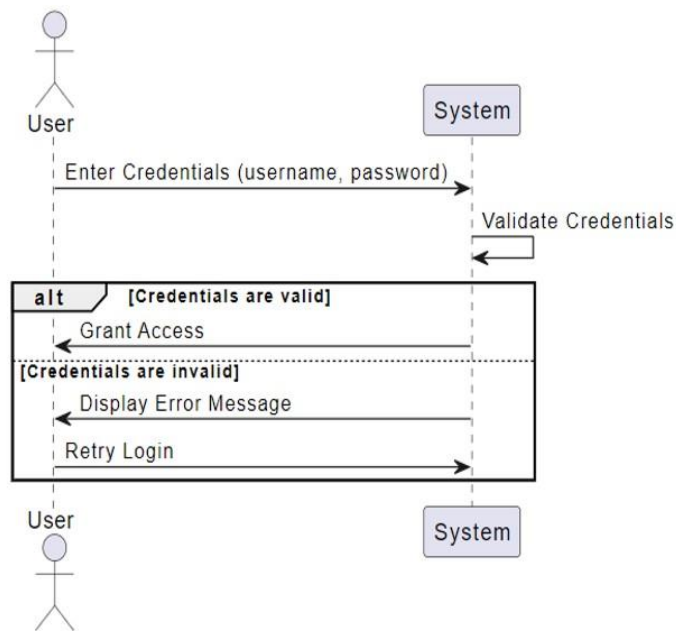


Figure 3.13: User Login Process Sequence

The login process includes:

- User requests login
- Credentials required
- User provides credentials
- Validation of credentials using database
- User receives login successful or failure message

#### 3.2.4.3 AI Image Generation Process

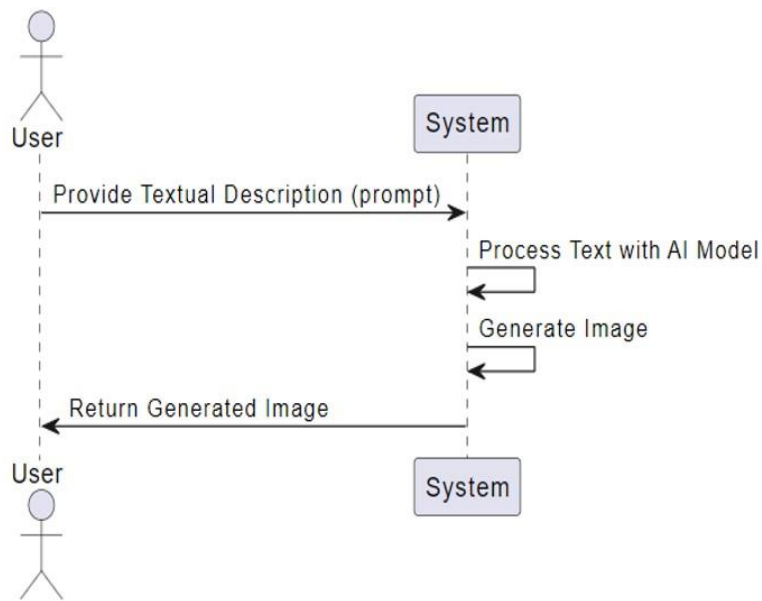


Figure 3.14: AI Image Generation Process Sequence

The AI image generation process involves:

- User inputs a text prompt
- Image Generation System processes the text prompt
- System uses an AI model to generate an image based on the prompt
- Generated image is returned and displayed to the user

## 3.2.4.4 Image Upload Process

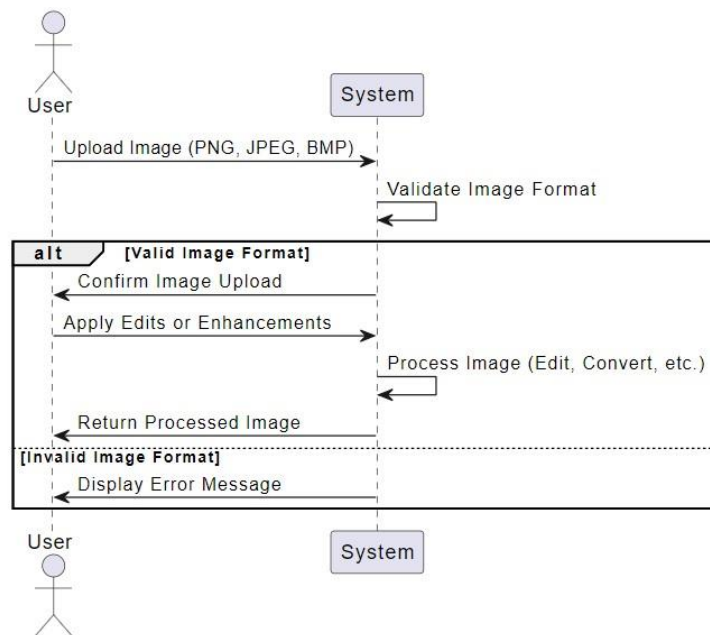


Figure 3.15: Image Upload Process Sequence

The image upload process consists of:

- User uploads their own image
- Chatbot System validates the image format
- System confirms successful upload
- Uploaded image is displayed to the user

#### 3.2.4.5 Background/Noise Removal Process

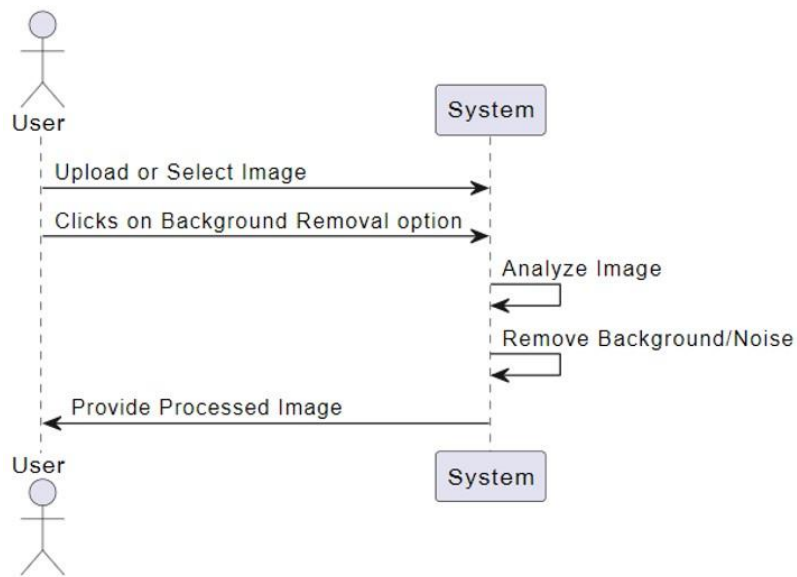


Figure 3.16: Background/Noise Removal Process Sequence

The background/noise removal process includes:

- User requests to remove background/noise from an image
- System processes the image to remove background/noise
- Processed image without background/noise is returned to the user



## 3.2.4.6 Image to SVG Conversion Process

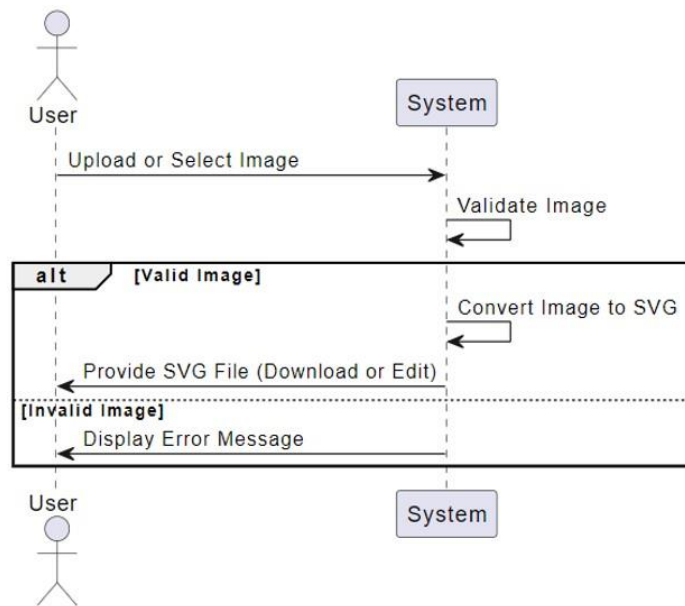


Figure 3.17: Image to SVG Conversion Process Sequence

The SVG conversion process consists of:

- User requests to convert an image to SVG format
- Image Processing System validates the image format
- System converts the image to SVG format
- Converted SVG image is returned to the user

#### 3.2.4.7 Image Download Process

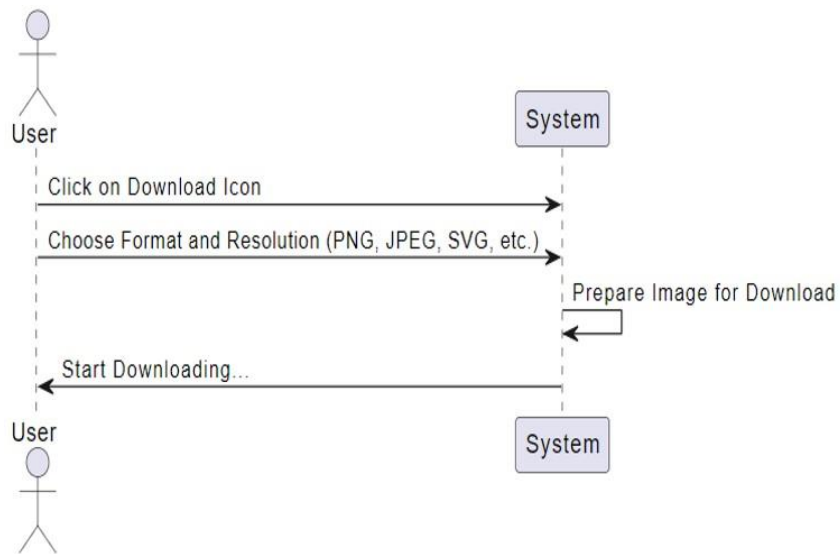


Figure 3.18: Image Download Process Sequence

The image download process involves:

- User clicks the download icon on an image
- Chatbot System prepares the image for download
- System provides a download link for the image
- User confirms the download
- System initiates the image download

## 3.2.4.8 Chat History Management Process

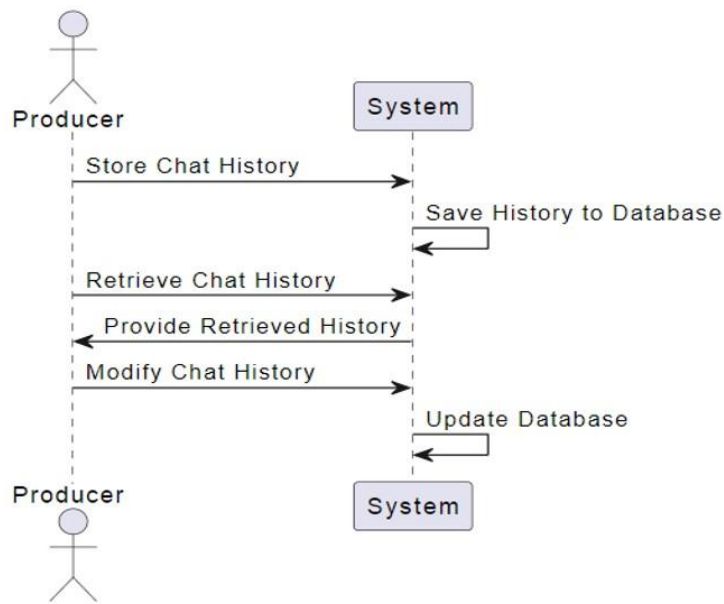


Figure 3.19: Chat History Management Process Sequence

The chat history management process includes:

- Producer requests to view chat history
- Chatbot System retrieves and displays chat history
- Producer selects a specific chat
- System fetches and displays selected chat details
- Producer can clear chat history
- System deletes history from database and confirms clearance

## 3.3 Data Design

### 3.3.1 Data Flow Diagram

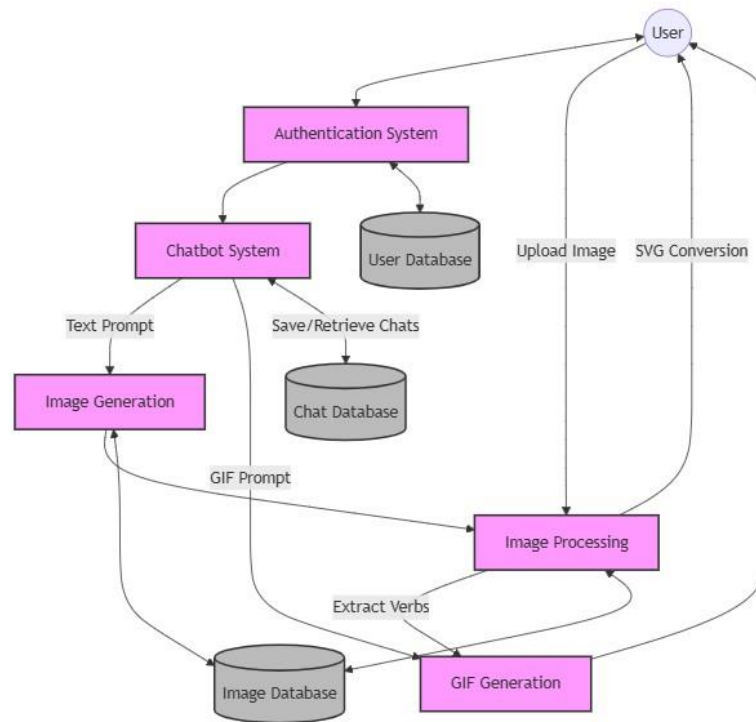


Figure 3.20: Data Flow Diagram

The Data Flow Diagram (DFD) illustrates the flow of information through our chatbot system. It shows the key processes, data stores, and how data moves between different components. Here's a breakdown of the main elements:

- **User:** The central entity interacting with the system.
- **Authentication System:** Manages user login and registration.
- **Chatbot System:** The core system that processes user inputs and manages interactions.
- **Image Generation:** Generates images based on text prompts.
- **Image Processing:** Handles image uploads, SVG conversion, and other imagerelated tasks.
- **GIF Generation:** Creates GIFs based on extracted verbs from prompts or images.
- **Data Stores:**
  - **User Database:** Stores user information
  - **Chat Database:** Stores conversation history
  - **Image Database:** Stores generated and uploaded images

Key data flows include:

- Text prompts from User to Chatbot System for image generation
- Image uploads and SVG conversion requests from User to Image Processing
- Auto generation of gif prompts or user define prompts for gif
- Retrieval of chat and images from the Database

This DFD is showcasing a high-level view that how data moves through the whole system, also highlighting the inter-connections between different components and databases.

## Conclusion

In conclusion, the Story Animator project will enable reversing the problems animators, designers, and other artists encounter when designing and modifying their animated tales. From a unique set of features, the platform simplifies character design through the use of image-to-SVG converter, background remover, GIF maker, and an AI character creator. In addition to saving time and money, these features promote innovation and creativity among a range of user classes, including content producers and professional animators. Scalability, security, and a seamless user experience are guaranteed by the strong system architecture, which also features a user-friendly interface and dependable data management. Story Animator has the potential to revolutionize the animation and design fields by providing high-quality tools to a wider audience, facilitating easier access to animation, and affordability.

# Bibliography

1. RunwayML, "Stable Diffusion v1.5," *RunwayML Documentation*  
Link: <https://runwayml.com>
2. Stability AI, "Stable Diffusion 2-1 Model," *Stability AI Documentation*  
Link: <https://stability.ai>
3. PlaygroundAI, "Playground V2.5 Release Notes," *Playground AI Documentation*  
Link: <https://playgroundai.com>
4. OpenCV, "Contours in OpenCV," *OpenCV Documentation Link:*  
<https://docs.opencv.org>
5. BRIA Background Removal, "BRIA Background Removal Tool," *BRIA Tech Overview Link:* <https://bria.com/background-removal>
6. Python Imaging Library (PIL), "Pillow: Python Imaging Library," *Pillow Documentation Link:* <https://pillow.readthedocs.io>
7. Hugging Face, "Transformers for Text-to-Image Generation," *Hugging Face Documentation Link:* <https://huggingface.co>